DrinkLens: Recurring Visual Feature Discovery from In-Situ Drinking Context Photos for Alcohol Use Prediction

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Abstract

Alcohol misuse remains a major public health concern, yet monitoring alcohol use in daily life is challenging because social, behavioral, and environmental contexts are often heterogeneous and can all influence an individual's drinking decisions. Existing studies on alcohol misuse largely rely on Ecological Momentary Assessment (EMA; e.g. self-report mood), wearable and mobile sensing (e.g., physiological, location, and vision-based sensors), and then use predefined features extracted from these data for drinking detection and estimated blood alcohol concentration (eBAC) prediction. However, these predefined features are manually selected and may miss implicit yet useful features that recur across heterogeneous day-to-day drinking contexts.

To address this gap, we present *DrinkLens*, a recurring visual feature discovery system for alcohol use prediction from in-situ drinking context photos. Photos are well suited for feature discovery because they capture rich visual details about the surrounding environment, social setting, activities, and objects. However, discovering meaningful contextual features from photos is challenging because photos vary substantially across individuals in settings and viewpoints. To address the challenge, *DrinkLens* first identifies recurring visual patterns across diverse drinking episodes through multi-granularity clustering and uses language-guided visual semantic abstraction to transform heterogeneous visual embeddings into explicit and semantically structured features. We then develop a context-aware eBAC prediction model that extracts the discovered visual features from new photos and integrates them with EMA and predefined variables.

We evaluate *DrinkLens* using real-world data collected from 48 participants. *DrinkLens* discovers 31 meaningful contextual features across 6 dimensions from the photos associated with daily alcohol use, and incorporating the new features into eBAC prediction reduces average absolute prediction error by 27.8% compared with models using only EMA and predefined variables.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; • **Applied computing** → **Health informatics**.

Keywords

Alcohol Use, Multimodal Sensing, Visual Feature Discovery

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1 Introduction

Alcohol misuse remains a major public health concern and contributes to substantial health risks, including injury, chronic disease, and premature mortality [8]. Monitoring alcohol use in daily life is particularly challenging because drinking episodes occur across heterogeneous social, behavioral, and environmental contexts [7]. These contexts, such as who a person is drinking with, what they are doing, and the surrounding environment, can contribute significantly to drinking behaviors, but are difficult to characterize outside controlled laboratory settings due to their high variability across everyday drinking episodes [5, 6]. Improving daily-life alcohol use prediction, therefore, requires a better understanding of the contexts in which drinking occurs.

Prior work has largely relied on ecological momentary assessment (EMA; e.g., self-reports of mood, environment, and alcohol use) and wearable and mobile sensing (e.g., physiological, behavioral, spatial, and visual modalities) to monitor alcohol use and capture its surrounding context in naturalistic settings [2, 3, 17]. Combined with machine learning models that use predefined features extracted from these data have improved the ecological validity of alcohol research and supported estimated blood alcohol

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concentration (eBAC) prediction [5, 15, 19]. However, they typically characterize drinking contexts using a fixed set of manually select variables [2, 4, 22]. Such predefined variables may overlook implicit yet useful features that recur across day-to-day drinking episodes.

In-situ drinking context photos capture rich visual information that had traditionally been used to derive predefined contextual variables. Building on this opportunity, we present *DrinkLens*, a recurring visual feature discovery system for alcohol use prediction and context understanding from in-situ drinking context photos. However, visual feature discovery is challenging because photos vary substantially across individuals in setting and viewpoint and often contain moment-specific details, such as a temporarily placed object on the table or a passerby in the background. To address this challenge, *DrinkLens* focuses on recurring visual patterns rather than isolated visual details. Our key insight is that visual features recurring across diverse drinking episodes are more likely to reflect meaningful drinking contexts than moment-specific visual details [2, 5]. Based on this insight, *DrinkLens* clusters in-situ drinking context photos at multiple resolutions, and develops a language-guided visual semantic abstraction pipeline that converts heterogeneous visual observations into explicit, semantically structured representations. These representations enable the discovery of recurring visual features that characterize common drinking contexts. Rather than assuming that these recurring visual contexts are causally related to alcohol consumption, we treat them as candidate contextual features for more in-depth alcohol use understanding and evaluate whether they improved eBAC prediction beyond EMA and predefined contextual variables.

This paper makes the following contributions:

- We develop *DrinkLens*, a system for discovering recurring contextual features from in-situ photos to characterize common yet implicit drinking contexts.
- We introduce a language-guided visual semantic abstraction pipeline that converts heterogeneous visual observations into explicit, semantically comparable representations at multiple levels of granularity.
- We evaluate the discovered 31 contextual features for alcohol use prediction and demonstrate their added predictive value beyond EMA and predefined variables.

Our evaluation shows that newly discovered visual-context features (e.g., screen-focused settings, documentary-style selfies, and room brightness) improve eBAC prediction beyond EMA and predefined variables. Incorporating these features reduces the average absolute prediction error from 0.0221 to 0.0160 eBAC, corresponding to a 27.8% reduction.

2 *DrinkLens* Framework

DrinkLens consists of two stages: Recurring-Context Discovery, Feature Extraction and Prediction. In the first stage, *DrinkLens* discovers a recurring context features list from recurring visual patterns using in-situ drinking photos. Then, in the second stage, *DrinkLens* uses this list to extract structured feature values from new episode photos and integrate them with EMA and predefined visual features for downstream eBAC prediction.

2.1 Stage I: Recurring Drinking Context Discovery from In-Situ Photos

2.1.1 Embedding In-Situ Drinking Context Photos. Given a set of in-situ photographs, we first map each photo into a visual embedding space through a local vision encoder.

$$\mathbf{z}_i = g_{\text{vis}}(I_i), \tag{1}$$

where I_i is the i-th image and $g_{\text{vis}}(\cdot)$ denotes the visual encoder. We then apply L_2 normalization. The normalized embeddings provide a shared representation for measuring visual similarity across photos. This allows the system to group photos based on scene structure, objects, people, lighting, and other visual-context cues without requiring predefined labels at the clustering stage.

2.1.2 Multi-Granularity Clustering. To discover recurring contextual features from drinking-episode photos, *DrinkLens* first needs to identify visual patterns that appear repeatedly across the dataset. We identify the recurrence through clustering in the photo embedding space. If multiple photos are visually similar, they would be grouped together and the cluster provides evidence of recurring visual context, such as a shared setting, object arrangement etc. In this sense, clustering is not used as a final label assignment, but as a discovery mechanism for finding repeated visual structures that may contain implicit contextual features.

However, the recurring context can exist on different levels of visual information from coarse-to-fine. Recurring context can appear in a broader context, as general drinking environment, or they share a finer pattern, as a specific details about the environment such as table or screen. Therefore, a fixed clustering value would be limited. Clustering reveals repeated visual patterns, but semantic level of the recurrence depends on clustering granularity.

This motivates multi-granularity recurring context formation. A smaller value of K produces broader clusters that capture high-level recurring contexts, while a larger value of K separates the embedding space into finer groups that may reveal more specific recurring cues. To select proper value of clustering, *DrinkLens* uses two complementary strategies: *data-driven clustering*, which estimates natural cluster structure from the embedding space, and *controlled granularity*, which applies pre-selected K values to explicitly examine coarse-to-fine levels of recurrence.

Data-driven clustering. This strategy allows the photo collection itself to suggest candidate clustering granularity. We estimate natural groupings in the CLIP embedding space using a density-based watershed procedure. Intuitively, dense regions in the embedding space indicate groups of photos that repeatedly share similar visual structure. For each normalized embedding $\tilde{\mathbf{z}}_i$, we compute a local density score:

$$\rho_i = \left(\frac{1}{k} \sum_{\tilde{\mathbf{z}}_j \in \mathcal{N}_k(i)} \left\| \tilde{\mathbf{z}}_i - \tilde{\mathbf{z}}_j \right\|_2 \right)^{-1}, \tag{2}$$

where $\mathcal{N}_k(i)$ is the set of k nearest neighbors of $\tilde{\mathbf{z}}_i$. The watershed procedure identifies density basins and uses the resulting structure to produce data-driven candidate values of K. This enables *DrinkLens* to follow naturally emerging recurring contexts in the photo embedding space.

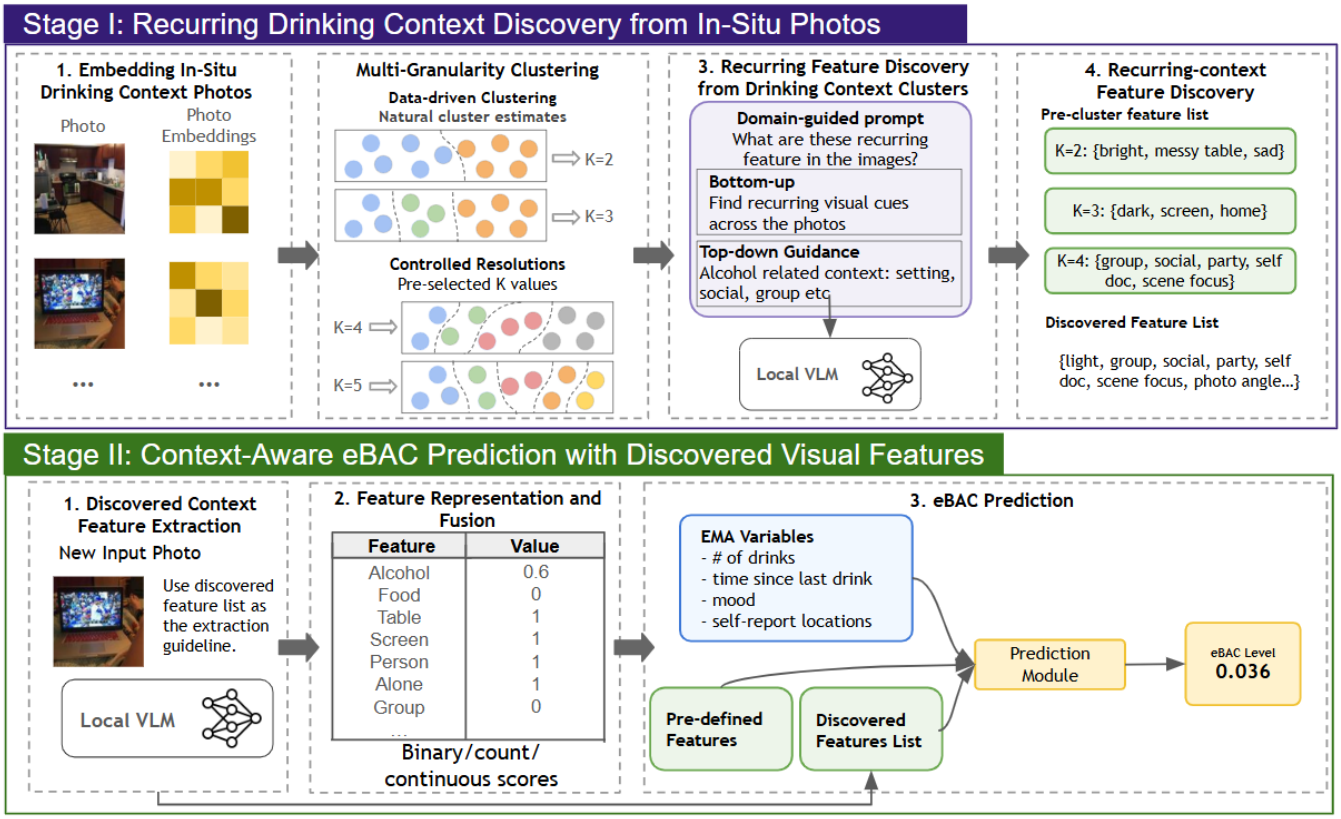


Figure 1: Overview of *DrinkLens*. Stage 1, *Recurring-Context Discovery*, clusters in-situ drinking context photos at multiple resolutions and uses language-guided visual semantic abstraction to produce structured representations, enabling discovery of recurring visual-context features not specified in advance. Stage 2, *Feature Extraction and Prediction*, applies the discovered feature list to individual photos, encodes structured photo-derived features, and combines them with EMA variables and predefined, human-verified image features for eBAC prediction.

Controlled Granularity. This strategy applies a small set of pre-selected K values to ensure systematic coverage across different levels of visual abstraction. Lower values of K capture broad recurring contexts, such as indoor dining, home, restaurant, or outdoor scenes, while higher values of K can reveal finer recurring cues, such as crowded tables, group selfies, low-light settings, food-and-drink arrangements, or screen-centered compositions. This prevents the discovery process from depending only on a single data-driven granularity and allows *DrinkLens* to search for recurring context from coarse to fine granularity.

2.1.3 Recurring Feature Discovery from Drinking Context Clusters. After multi-granularity clustering, each cluster provides evidence that a visual pattern recurs across multiple drinking-episode photos. However, the cluster assignment itself does not explain what contextual cues are shared by the photos. Therefore, *DrinkLens* uses a local vision-language model (VLM) to interpret each cluster into semantically structured recurring-context features. For each cluster, *DrinkLens* selects representative photos closest to the cluster center in the embedding space. These representative photos are provided to the VLM together with a recurring-context discovery prompt. The prompt is designed with two complementary components:

Bottom-up discovery. The VLM compares representative photos from the same cluster and identifies visual elements that repeatedly appear across the examples. This encourages the model to focus on cluster-level recurrence rather than isolated details from a single image, allowing *DrinkLens* to discover cluster-specific patterns without relying only on a fixed feature vocabulary.

Top-down guided discovery. The VLM is also provided with reference context categories related to alcohol use and eBAC prediction, including location, social setting, environment, food, visible drinks, lighting, activity context, and camera viewpoint. These categories guide the model toward domain-relevant contextual signals, while the prompt requires each extracted feature to be supported by visual evidence from the representative photos.

Together, these two components allow the VLM to identify features that are both grounded in repeated visual patterns and aligned with alcohol-relevant context. To reduce unsupported inference, the prompt restricts the model to features that are visually observable, appear across multiple representative photos, and can be expressed as structured variables, such as binary, categorical, count-based, or ordinal features. For each cluster, the VLM returns a structured recurring-context record containing a short cluster summary, recurring visual cues, candidate feature tags, feature

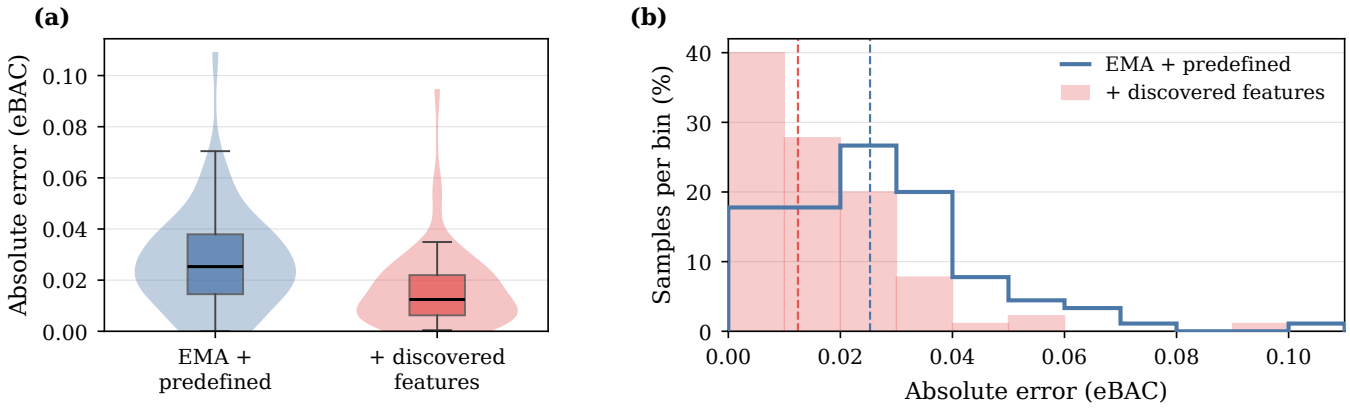


Figure 2: Prediction error with discovered visual-context features. (a) Absolute eBAC error for EMA + predefined variables and the model with discovered features. (b) Error histograms for the two models. Dashed lines show median error.

types, and supporting visual evidence. The candidate features from all clusters and clustering granularity are then consolidated into a shared discovered feature list for downstream feature extraction.

2.1.4 Recurring-context Feature List Construction. The recurring-context records from all clusters and clustering granularity are merged into a discovered feature list. To reduce duplicate features, *DrinkLens* maintains a running list of previously discovered feature tags and prompts the VLM to reuse an existing tag when a new cluster contains the same one. A new feature is added only when it represents a visually distinct recurring pattern. This simple consolidation step produces a shared feature vocabulary for downstream extraction while avoiding repeated or semantically identical tags.

2.2 Stage II: Context-Aware eBAC Prediction with Discovered Visual Features

This stage uses the discovered feature list as a feature extraction guide, combines the extracted photo-derived features with EMA variables and predefined visual features, and trains a regression model to predict eBAC.

2.2.1 Discovered Context Feature Extraction. Given a new photo, *DrinkLens* prompts the local VLM with the discovered recurring-context feature list and asks the model to determine which features are visually present in the image. This step differs from the discovery stage. In Stage I, the VLM identifies candidate features by comparing representative photos within recurring visual clusters. In Stage II, the VLM uses the finalized feature list as a fixed vocabulary and extracts feature values from each individual photo.

2.2.2 Feature Representation and Fusion. The VLM output is encoded as structured variables that can be used by a tabular prediction model. Depending on the feature definition, discovered recurring-context features may be represented as binary, categorical, count-based, or ordinal variables. For example, `food_visible` may be encoded as a binary feature, `people_count` as a count-based feature, and `lighting_condition` as a categorical feature. Each feature is stored together with its observability indicator and confidence score.

For each prediction sample, *DrinkLens* combines three groups of predictors. First, EMA captures participant-reported drinking and contextual information, such as alcohol use, time, location, and mood. Second, predefined visual features capture researcher-specified image variables. These predefined features are not selected by the VLM; they are extracted using established object-detection or image-processing methods and manually verified when needed. Third, discovered recurring-context features are extracted using the feature-guided VLM procedure described above.

2.2.3 eBAC Prediction. *DrinkLens* combines EMA variables, predefined visual features, discovered recurring-context features, and observability coverage scores into a feature vector:

$$\mathbf{x}_{p,t} = \left[\mathbf{x}_{p,t}^{\text{EMA}} \quad \mathbf{x}_{p,t}^{\text{pre}} \quad \mathbf{x}_{p,t}^{\text{disc}} \quad \mathbf{q}_{p,t}\right]. \tag{3}$$

The prediction model estimates eBAC as:

$$\hat{y}_{p,t} = f_\theta(\mathbf{x}_{p,t}), \tag{4}$$

where f_θ is a regression model and $\hat{y}_{p,t}$ is the predicted eBAC value.

3 Real-World Evaluation

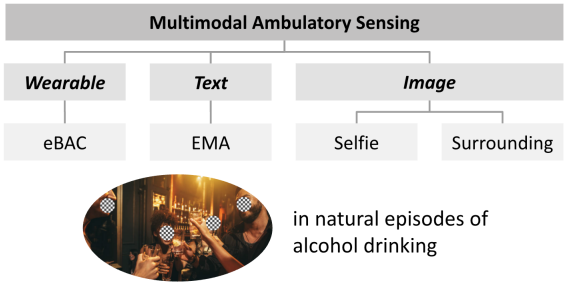


Figure 3: Overview of the multimodal data collection in natural drinking episodes.

3.1 Multimodal Sensing Data Collection

We developed a multimodal sensing system to capture alcohol use and surrounding drinking contexts in everyday life, as shown in Figure 3. Participants completed one week of intensive mobile sensing, wearing a transdermal alcohol ankle monitor for continuous alcohol assessment and responding to six random smartphone prompts per day. At each prompt, participants reported their current mood and took photographs of their immediate environment; approximately

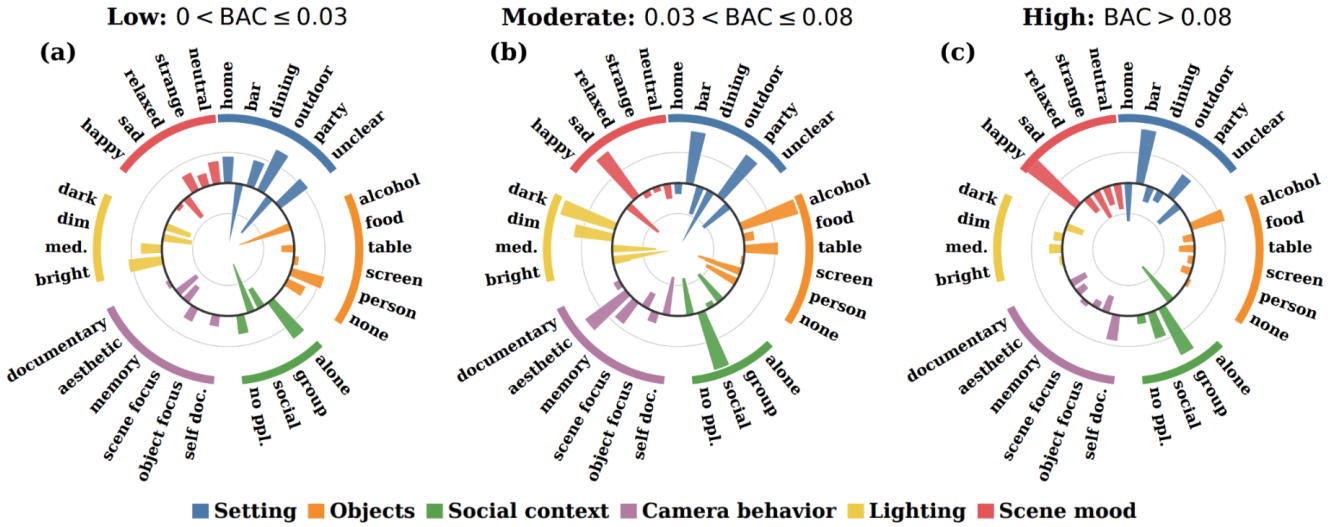


Figure 4: Visual-context feature correlations across low, moderate, and high BAC categories. Features are grouped by setting, objects, social context, camera behavior, lighting, and scene mood.

half of the participants also provided selfies. Participants also supplied brief captions describing the scene and daily self-reports of alcohol consumption, which were used as secondary validation checks. Alcohol consumption was estimated from transdermal alcohol concentration readings collected by the biosensor.

3.2 Alcohol Use Prediction Results

We first evaluate whether discovered visual-context features improve eBAC prediction beyond EMA and predefined variables. As shown in Figure 2(a), adding discovered features reduces the average absolute error from 0.0221 to 0.0160 eBAC, corresponding to a 27.8% reduction. This reduction is practically meaningful because a change of approximately 0.02 BAC is often used as a rough reference for the effect of one standard drink, although the exact change depends on individual factors such as body weight, sex, drinking pace, and metabolism. Reducing the average error below this range suggests that the model is better able to capture smaller differences in intoxication level.

Figure 2(b) further shows that the model with discovered features has more predictions concentrated in the low-error range. Together, these results suggest that participant photos provide complementary contextual information that is not fully captured by EMA responses or predefined variables. For example, visual features may encode setting, alcohol-related objects, social context, lighting, and scene mood around drinking episodes. Thus, the discovered features improve prediction by incorporating contextual cues that are difficult to manually specify in advance.

3.3 Feature Discovery Results and Analysis

To interpret the discovered features, we examine their relative correlation across BAC categories. Figure 4 summarizes feature correlations in low, moderate, and high BAC samples. The features are grouped into six interpretable dimensions based on their meaning: setting, objects, social context, camera behavior, lighting, and scene mood. Moderate and high BAC samples show stronger correlation in drinking-related objects, social context, and certain indoor or social settings, suggesting that the discovered features capture meaningful context around drinking occasions.

These patterns help explain why the discovered features improve prediction performance. Object- and setting-related cues, such as alcohol, food, table, and indoor-scene patterns, provide visual evidence of the immediate drinking environment. Composition-related cues, such as screen-centered, object-focused, or scene-focused photos, describe what participants were attending to or documenting during the episode. Self-documentation and person-related cues may further reflect social context or differences in photo-taking behavior, while lighting-related cues distinguish environmental conditions such as dim or bright settings. Compared with EMA, photos can provide additional evidence about the environment and activity context at the time of drinking. The observed associations may also reflect participant composition, photo-taking behavior, or the smaller sample size in the high-BAC category.

4 Related Work

This section reviews prior work on alcohol use monitoring, photo-based context understanding, and visual context discovery.

4.1 Alcohol Use Monitoring in Daily-Life

Alcohol use occurs within heterogeneous social, behavioral, and environmental contexts that are difficult to comprehensively capture in daily life [1, 6, 7]. Ecological momentary assessment (EMA) has been widely used to capture drinking behavior, mood, craving, and social setting near the time of drinking [17], while estimated blood alcohol concentration (eBAC) provides a measure of alcohol exposure when direct BAC measurement is unavailable or infeasible [9]. Mobile and wearable sensing further support passive and continuous alcohol monitoring. Smartphone sensing has used location, accelerometer, phone usage to detect and characterize drinking-related behavior [3, 4, 13, 21], while transdermal alcohol

sensors continuously measure transdermal alcohol concentration (TAC) [11]. Multimodal ambulatory studies have further combined EMA, wearable alcohol sensing, and in-situ photographs to jointly characterize drinking contexts and alcohol use [2, 5, 19, 22]. However, context is typically represented through predefined variables, potentially missing implicit yet useful contextual features.

4.2 Photo-Based Context Understanding

Photographs provide a complementary source of contextual information by capturing visual details of environments, activities, objects, and social settings. In health sensing, camera-based approaches have supported mobile food logging, dietary assessment, and activity recognition from egocentric and wearable cameras [10, 14, 18]. These studies demonstrate the potential of images to capture aspects of daily-life context that are difficult to obtain through self-report or conventional sensing alone. Alcohol studies have similarly collected in-situ photographs with EMA, and extracted information about physical setting, activity, interpersonal context, and drinking environments [2, 5].

4.3 Visual Context Discovery

Recent vision-language models (VLMs), including CLIP, LLaVA, and GPT-4V, provide new opportunities to extract semantic information from images through vision-language representations and natural-language descriptions. In parallel, clustering has been used to identify repeated structure in daily-life image collections, such as grouping egocentric images into contextually similar events for dietary assessment [16]. Language-guided methods further cluster images using generated textual descriptions [20], while open-ended approaches automatically discover interpretable grouping criteria from unstructured image collections [12]. Together, these studies demonstrate the potential of VLM reasoning and clustering to uncover interpretable structure in visual data. However, their primary goals are typically to organize image collections, discover semantic concepts, or explain visual groupings, rather than identify recurring visual features for daily-life health prediction.

5 Conclusion

We presented *DrinkLens*, a recurring visual feature discovery system for alcohol use prediction and context understanding from in-situ drinking context photos. Through cluster-guided feature discovery and local VLM-based extraction, *DrinkLens* converts unstructured photos into structured visual-context features. Our results show that these features reduce average absolute eBAC prediction error from 0.0221 to 0.0160. Future work will improve the interpretability of discovered visual features by linking them to theories of alcohol use and modeling their relationships.

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